

Learning as a shortcut to gravitational-wave inference

A neural approximation to an expensive likelihood could make gravitational wave analysis faster and more adaptable.

Gravitational waves (GW) are ripples in spacetime intrigued by movement of massive objects such as black holes or neutron stars. Decade after the first observation of GW plenties of signals are thereafter recorded, meanwhile more complicated theoretical models have been put forward to investigate more phenomona even the beyond general relativity ones. All these work relies on proper parameter estimation (PE) which is both highly demanding in computational resources and poor in reusability when changing any part of PE. Negri and Samajdar¹ develop an algorithm featuring neural likelihood estimation to significantly accelerate PE and for the first time apply such algorithm on a real GW signal.

Gravitational waves carry information about their sources, but detectors record a noisy strain data rather than the parameters themselves including mass, angular momentum and direction etc. Recovering these properties requires comparing the data with theoretical waveforms determined by them, acquiring probability distributions of parameters.

Such recovery process is Bayesian Inference, the information of parameters of interest is encoded in the posterior probability density function:

$$p(\vec{\theta} | H_s, d) = \frac{\mathcal{L}(d | \vec{\theta}, H_s) p(\vec{\theta} | H_s)}{\mathcal{Z}}. \quad (1)$$

Here d is for the exact data recorded by detectors, $\vec{\theta}$ for parameters of interest, H_s stands for the hypothesis that data contains GW signal, and $\mathcal{Z}$ is called evidence that signifies support of the hypothesis. The likelihood $\mathcal{L}$ measures how well the data is explained by proposed parameters. In the GW detection case, it is calculated indirectly under the assumption that noise adhere to Gaussian distribution. This process includes cumbersome integrals and is challenging in computation. Additionally the high dimensionality of parameters requires $\mathcal{O}(10^6 \sim 10^8)$ likelihood points to depict the whole distribution. These factors together make the conventional Bayesian PE computationally expensive.²

Significant progress has been made in faster and cheaper PE, existing approaches for example features by skipping the burden of likelihood evaluation by a likelihood-free inference, training neural network to estimate posterior directly, reducing the computational costs in evaluating likelihood directly.^{3;4;6} Their differing routes of reaching posterior, training requirements or validation strategies leave room for complementary methods. Some approaches does not retain true likelihood, while some others leave evidence unknown.

The core concept of Negri and Samajdar¹'s PE algorithm is incorporating neural likelihood estimator in conventional Bayesian inference algorithm. The idea is based on neural network (NN)'s ability of approximating any mapping. An NN is trained as a mapping $[\vec{\theta}_i] \rightarrow [\mathcal{L}(\vec{\theta}_i)]$ that resembles the true likelihood function based on selected training points of true likelihood evaluation(Fig.1a). Once trained, NN estimates likelihood values through a short sequence of weighted sums and simple nonlinear operations, bypassing the expensive true likelihood evaluation. In total NN's cost of training can be paid off by cheap evaluation of likelihood via itself during extensive sampling needed in PE, provided that the NN approximation remains sufficiently accurate.

Afterwards, NN could fit in Bayesian PE as replacement of true likelihood function. The posterior is then acquired by stochastic sampling across parameter space for NN likelihood estimation and reviewed by a metric to assess its accuracy in approximating true posterior. If examination failed, a new training cycle is restarted, with sampled posterior points and their true likelihood being added into the training set. This allows the network to correct itself from inaccurate regions. Fig.1b illustrates the improvement: a spurious peak at low mass ratio in the first cycle disappears

with further training, while the recovered distributions approach those obtained using the true likelihood.

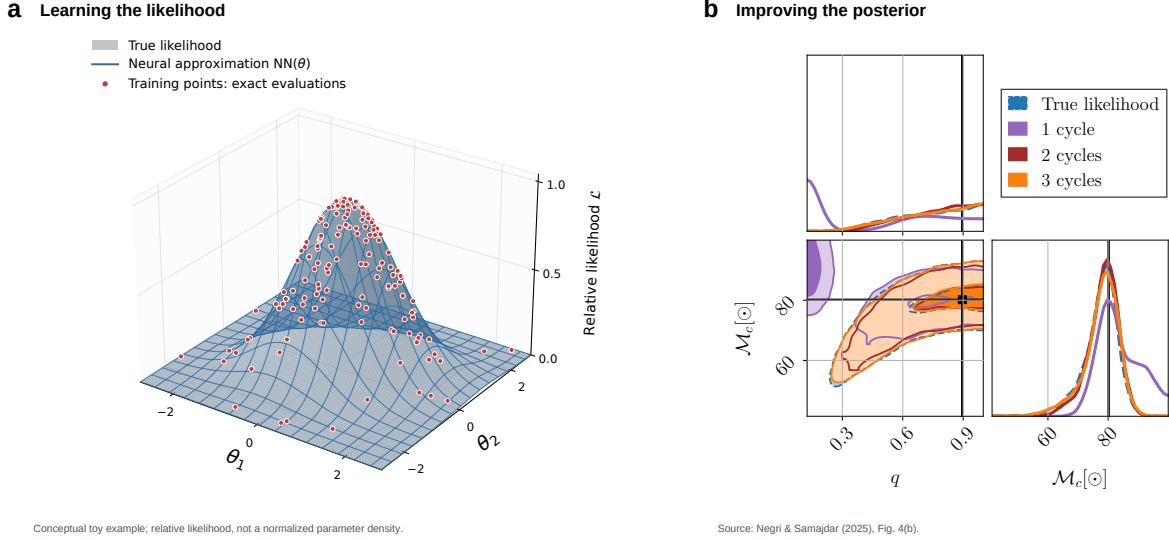


Figure 1: Learning and refining a likelihood approximation. **a**, In a conceptual two-parameter example, training set consists of true likelihood evaluations (red dots) and constrain a neural network approximation (blue net surface) to a true likelihood surface (grey volume). Points of training set concentrate near the likelihood peak while also covering the outskirts; the height is relative likelihood. **b**, Successive training cycles improve posterior agreement with a reference analysis for parameters mass ratio and chirp mass. Subfigure b is reproduced from Fig. 4(b) of Negri and Samajdar.¹

Negri and Samajdar¹ first test their method on 99 simulated binary black hole signals. They check whether the inferred uncertainty ranges of parameters contained these known parameter values at the expected rates: a range assigned 90% probability should contain the true value in roughly 90% of the simulations. The overall agreement among all parameters supports the reliability of uncertainty estimates. They then analyze GW150914, a real signal from a black hole merger. Using the same waveform model, the neural likelihood recovers parameter distributions consistent with a conventional analysis while reducing the reported total CPU time, including training, from almost eleven hours to forty minutes. Analyses using four different waveform models also yield highly consistent source properties. These results demonstrate substantial computational savings within the range of signals tested.

Neural likelihood estimator is successful in approximating posterior in simulations and GW150914, it features in being adaptable to various types of likelihood and keeping evidences like conventional Bayesian PE. Nevertheless it comes with limitations in many aspects. First is that the validation is implemented under simplified case of 9-parameter gravitational wave source. For example binaries whose spinning black holes carry non-parallel orbital and spin momentum would be more demanding in parameter dimensionality. Louder signals also weakens samplers' performance. Even the method in preparing training set is left for refinement in better approximating posterior.

With further validation, cheaper inference by virtue of neural network could accelerate the examination of same events using a wider range of waveform models and physical assumptions on the fly. Such comparisons matter when researchers investigate how black hole binaries form or search for departures from general relativity. Agreement across analyses would strengthen confidence in these conclusions; disagreements could expose limitations in the models or even clues interpreted as new physics. Neural likelihood estimation could therefore help turn growing collections of signals into more thoroughly tested accounts of the systems that produced them.

AI Usage Statement

AI tools are used in:

1. Grammar and spelling check
2. Illustration of Fig.1

References

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